

EX.NO:3 INHERITANCE PROGRAM FOR AN EMPLOYEE CLASS

PROGRAM:

```
import java.util.Scanner;
class Vehicle {
    String vehicleNumber;
    String model;
    String manufacturer;
    double price;
    Vehicle(String vehicleNumber, String model, String manufacturer, double price) {
        this.vehicleNumber = vehicleNumber;
        this.model = model;
        this.manufacturer = manufacturer;
        this.price = price;
    }
    void display() {
        System.out.println("Vehicle Number : " + vehicleNumber);
        System.out.println("Model : " + model);
        System.out.println("Manufacturer : " + manufacturer);
        System.out.println("Price : " + price);
    }
}
class Car extends Vehicle {
    Car(String v, String m, String man, double p) {
        super(v, m, man, p);
    }
    void bill() {
        double tax = price * 0.05;
        double insurance = 10000;
        double total = price + tax + insurance;

        System.out.println("\n***** CAR BILL *****");
        display();
        System.out.println("Road Tax : " + tax);
        System.out.println("Insurance : " + insurance);
        System.out.println("Total Cost : " + total);
    }
}
class Bike extends Vehicle {
    Bike(String v, String m, String man, double p) {
        super(v, m, man, p);
    }
    void bill() {
        double tax = price * 0.03;
        double insurance = 3000;
        double total = price + tax + insurance;
```

```

        System.out.println("\n***** BIKE BILL *****");
        display();
        System.out.println("Road Tax : " + tax);
        System.out.println("Insurance : " + insurance);
        System.out.println("Total Cost : " + total);
    }
}

```

```

class Truck extends Vehicle {
    Truck(String v, String m, String man, double p) {
        super(v, m, man, p);
    }
    void bill() {
        double tax = price * 0.10;
        double insurance = 15000;
        double total = price + tax + insurance;

        System.out.println("\n***** TRUCK BILL *****");
        display();
        System.out.println("Road Tax : " + tax);
        System.out.println("Insurance : " + insurance);
        System.out.println("Total Cost : " + total);
    }
}

```

```

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Vehicle Number: ");
        String number = sc.nextLine();

        System.out.print("Enter Model: ");
        String model = sc.nextLine();

        System.out.print("Enter Manufacturer: ");
        String manufacturer = sc.nextLine();

        System.out.print("Enter Vehicle Price: ");
        double price = sc.nextDouble();

        System.out.println("\n1. Car");
        System.out.println("2. Bike");
        System.out.println("3. Truck");
        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        switch (choice) {

```

```

    case 1:
        Car c = new Car(number, model, manufacturer, price);
        c.bill();
        break;

    case 2:
        Bike b = new Bike(number, model, manufacturer, price);
        b.bill();
        break;

    case 3:
        Truck t = new Truck(number, model, manufacturer, price);
        t.bill();
        break;

    default:
        System.out.println("Invalid Choice");
}

sc.close();
}
}

```

OUTPUT:

```

Enter Vehicle Number: TN32AB1234
Enter Model: Swift
Enter Manufacturer: Maruti
Enter Vehicle Price: 800000

```

```

1. Car
2. Bike
3. Truck
Enter Choice: 1

```

***** CAR BILL *****

```

Vehicle Number : TN32AB1234
Model          : Swift
Manufacturer   : Maruti

```

Price : 800000.0
Road Tax : 40000.0
Insurance : 10000.0
Total Cost : 850000.0

EX.NO:4 CREATE AN ABSTRACT CLASS – SHAPE CLASS

PROGRAM:

```
import java.util.Scanner;
abstract class LibraryMember {
    int memberId;
    String name, email, phone;
    LibraryMember(int memberId, String name, String email, String phone) {
        this.memberId = memberId;
        this.name = name;
        this.email = email;
        this.phone = phone;
    }
    void display() {
        System.out.println("Member ID : " + memberId);
        System.out.println("Name : " + name);
        System.out.println("Email : " + email);
        System.out.println("Phone : " + phone);
    }

    abstract void generateSummary();
}
class StudentMember extends LibraryMember {

    StudentMember(int id, String name, String email, String phone) {
        super(id, name, email, phone);
    }
    void generateSummary() {
        System.out.println("\n---- Student Member ----");
        display();
        System.out.println("Borrowing Limit : 5 Books");
        System.out.println("Penalty Per Day : Rs.2");
        System.out.println("Annual Membership Fee: Rs.200");
    }
}
class FacultyMember extends LibraryMember {

    FacultyMember(int id, String name, String email, String phone) {
```

```

        super(id, name, email, phone);
    }
    void generateSummary() {
        System.out.println("\n---- Faculty Member ----");
        display();
        System.out.println("Borrowing Limit    : 10 Books");
        System.out.println("Penalty Per Day    : Rs.3");
        System.out.println("Annual Membership Fee: Rs.500");
    }
}
class ExternalMember extends LibraryMember {

    ExternalMember(int id, String name, String email, String phone) {
        super(id, name, email, phone);
    }

    void generateSummary() {
        System.out.println("\n---- External Member ----");
        display();
        System.out.println("Borrowing Limit    : 3 Books");
        System.out.println("Penalty Per Day    : Rs.5");
        System.out.println("Annual Membership Fee: Rs.1000");
    }
}
public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Member ID: ");
        int id = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Email: ");
        String email = sc.nextLine();
        System.out.print("Enter Phone: ");
        String phone = sc.nextLine();
        System.out.println("\nSelect Member Type");
        System.out.println("1. Student");
        System.out.println("2. Faculty");
        System.out.println("3. External");
        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        switch (choice) {
            case 1:
                StudentMember s = new StudentMember(id, name, email, phone);
                s.generateSummary();

```

```

        break;
    case 2:
        FacultyMember f = new FacultyMember(id, name, email, phone);
        f.generateSummary();
        break;
    case 3:
        ExternalMember e = new ExternalMember(id, name, email, phone);
        e.generateSummary();
        break;
    default:
        System.out.println("Invalid Choice");
    }
    sc.close();
}
}

```

OUTPUT:

Enter Member ID: 101
 Enter Name: Janaki
 Enter Email: janaki@gmail.com
 Enter Phone: 9876543210

Select Member Type

1. Student
2. Faculty
3. External

Enter Choice: 1

---- Student Member ----

Member ID : 101
 Name : Janaki
 Email : janaki@gmail.com
 Phone : 9876543210
 Borrowing Limit : 5 Books
 Penalty Per Day : Rs.2
 Annual Membership Fee: Rs.200